How Does the ∼20% Increase in Post-Merger Gravitational-Wave Luminosity from Neutrino Flavor Transformation Affect Future Detection Prospects with Cosmic Explorer and the Einstein Telescope?

Seminar Response — Based on Qiu et al., Phys. Rev. Lett. **135**, 091401 (2025)

1 Framing the Problem: What Is the 20% Effect?

To understand why a 20% increase in gravitational-wave (GW) luminosity matters for future experiments, we first need to understand where it comes from physically and what it means in the language of detector physics.

In Qiu et al. (2025), the authors run three numerical relativity simulations of equal-mass ($1.35\,M_\odot$–$1.35\,M_\odot$) binary neutron star (BNS) mergers using the DD2 equation of state. The three cases differ only in their treatment of neutrino flavor conversion:

(i) **No-Mixing**: the standard approach in all prior merger simulations—neutrinos do not change flavor.

(ii) ρ–**11**: flavor equilibration is activated wherever the rest-mass density falls below 10^{11} g/cm^3 (the optically thin regime for all neutrino species).

(iii) ρ–**13**: flavor equilibration is activated below 10^{13} g/cm^3 (extending into the disk interior, where the matter is still optically thick to electron-type neutrinos but thin to heavy-lepton neutrinos).

The key result: the ρ–13 remnant radiates approximately 20% more total energy in gravitational waves over the ∼30 ms of post-merger evolution than the No-Mixing remnant. The ρ–11 case falls between the two.

1.1 The Physical Mechanism (for a Particle Physics Audience)

If you are accustomed to thinking about neutrinos as mass eigenstates propagating through vacuum, the relevant new ingredient here is the *neutrino self-interaction potential*. In the dense environment of the merger remnant, neutrino number densities are so large ($n_\nu \sim 10^{33}$ cm^{-3}) that the forward-scattering potential from neutrino-neutrino coherent scattering, mediated by Z^0 exchange in the t-channel, becomes comparable to or exceeds the matter

(MSW) potential. Schematically, the Hamiltonian for a neutrino of momentum $\mathbf{p}$ in the flavor basis contains a term:

$$H_{\nu\nu} \propto \sqrt{2}\, G_F \int \frac{d^3\mathbf{q}}{(2\pi)^3} \, (1 - \hat{\mathbf{p}} \cdot \hat{\mathbf{q}}) \, \big[\rho(\mathbf{q}) - \bar{\rho}(\mathbf{q})\big], \tag{1}$$

where $\rho(\mathbf{q})$ and $\bar{\rho}(\mathbf{q})$ are the neutrino and antineutrino density matrices in flavor space, and G_F is the Fermi constant. This potential is nonlinear—it depends on the flavor state of the entire neutrino ensemble—and can drive collective oscillations called *fast flavor instabilities* on nanosecond timescales.

Qiu et al. do not simulate these oscillations directly (that would require resolving ∼cm-scale coherence lengths inside a ∼100 km remnant). Instead, they model the *outcome* of such instabilities: a rapid relaxation toward flavor equilibrium, enforced through a Bhatnagar-Gross-Krook (BGK) collision operator of the form

$$\left.\frac{\partial f_\alpha}{\partial t}\right|_{\text{mixing}} = \frac{1}{\tau_a}\big(f_\alpha^{(a)} - f_\alpha\big), \tag{2}$$

with relaxation time $\tau_a = 0.4$ ns and the asymptotic state $f^{(a)}$ determined by the detailed-balance condition

$$\nu_e\bar{\nu}_e \rightleftharpoons \nu_\mu\bar{\nu}_\mu \rightleftharpoons \nu_\tau\bar{\nu}_\tau, \tag{3}$$

which, for the two-effective-species scheme (ν_e, ν_x where $x = \mu + \tau$), yields $4\, n_e^{(a)}\, n_{\bar{e}}^{(a)} = n_x^{(a)}\, n_{\bar{x}}^{(a)}$.

The physical consequence is that electron-type neutrinos and antineutrinos, which are copiously produced in the hot remnant via charged-current weak processes ($e^- + p \to \nu_e + n$, $e^+ + n \to \bar{\nu}_e + p$), are partially converted into heavy-lepton neutrinos (ν_μ, ν_τ and their antiparticles). Heavy-lepton neutrinos have much smaller interaction cross sections in this environment—they interact only via neutral currents and lack the charged-current channel available to ν_e. Therefore, converting electron neutrinos to heavy-lepton neutrinos *increases the net energy loss rate from the outer layers of the remnant*. The remnant loses thermal pressure support faster, contracts more, and the resulting denser, more compact configuration orbits at higher angular velocity and radiates gravitational waves more efficiently.

This is the origin of the ∼20% increase in GW luminosity: it is an indirect consequence of enhanced neutrino cooling driven by flavor transformation.

2 Gravitational-Wave Detection: A Quick Primer

For those unfamiliar with GW detector physics, here is the essential framework.

2.1 Strain Sensitivity and Signal-to-Noise Ratio

A GW detector measures the dimensionless strain $h(t) = \Delta L/L$, where ΔL is the differential arm-length change induced by a passing wave in an interferometer of arm length L. For a

source at luminosity distance D_L emitting GW power $\dot{E}_{\rm GW}$, the characteristic strain scales as

$$h \propto \frac{1}{D_L}\sqrt{\dot{E}_{\rm GW}}. \tag{4}$$

The detectability of a signal is quantified by the matched-filter signal-to-noise ratio (SNR):

$$\mathrm{SNR}^2 = 4\int_0^\infty \frac{|\tilde{h}(f)|^2}{S_n(f)}\,df, \tag{5}$$

where $\tilde{h}(f)$ is the Fourier transform of the GW strain and $S_n(f)$ is the one-sided noise power spectral density of the detector. For post-merger signals, the relevant frequency band is approximately $f \in [2, 4]$ kHz, since the dominant oscillation mode of the remnant (labeled f_2 in the literature) typically falls in this range.

A signal is conventionally considered "detected" if SNR $\gtrsim$ 5–8 for post-merger searches, and "well-characterized" (i.e., spectral features like f_2 can be measured) at SNR $\gtrsim$ 8–10.

2.2 Why Post-Merger Detection Is Hard

The post-merger GW signal is intrinsically difficult to detect for two reasons:

(a) Short duration: The remnant oscillates for only $\mathcal{O}$(10–100) ms before either settling into a stable configuration (with rapidly decreasing GW amplitude) or collapsing to a black hole. Short signals accumulate less SNR than the inspiral, which sweeps through the detector band over minutes to hours.

(b) High frequency: The f_2 peak falls at 2–4 kHz, where current detectors (Advanced LIGO/Virgo) are severely limited by photon shot noise. At these frequencies, the noise power spectral density $S_n(f)$ rises steeply, suppressing the integrand in Eq. (5). For reference, the post-merger signal from GW170817 (at $D_L \approx 40$ Mpc) was *not* detected; searches placed only upper limits, with estimates suggesting confident detection would have required a source distance of 1–5 Mpc with current instruments.

3 Cosmic Explorer and the Einstein Telescope

3.1 Cosmic Explorer (CE)

Cosmic Explorer is a proposed next-generation ground-based GW observatory in the United States, expected to become operational in the mid-2030s to 2040s. The baseline design calls for two facilities: one with 40 km arms and one with 20 km arms (compared to Advanced LIGO's 4 km arms). The sensitivity improvement is roughly an order of magnitude in strain amplitude across most of the detection band.

The 20 km facility is specifically motivated by post-merger science. At frequencies above ∼2 kHz, a longer arm can actually *degrade* sensitivity because the GW period becomes comparable to the light travel time down the arm ($L/c \approx 0.13$ ms for 40 km, corresponding to ∼7.5 kHz). The 20 km detector, potentially operated in a "tuned" configuration where quantum noise is optimized in the 2–4 kHz band at the expense of lower-frequency performance,

provides roughly 30% better average SNR in the post-merger band compared to the 40 km detector.

At 100 Mpc with a 40 km CE, post-merger SNRs for optimally oriented BNS mergers are expected to be $\mathcal{O}(5\text{–}15)$ depending on the equation of state and remnant lifetime. The BNS merger rate within 100 Mpc is approximately 2–3 per year at median estimates.

3.2 Einstein Telescope (ET)

The Einstein Telescope is a proposed European third-generation detector with a triangular underground configuration housing six interferometers (three pairs optimized for low and high frequency respectively), each with 10 km arms. Its design sensitivity exceeds current detectors by roughly a factor of 8–10 in amplitude. ET is expected to detect $\sim 10^5$ BNS coalescences per year across all redshifts, many with sufficient SNR to measure tidal deformability and, for nearby events, to probe the post-merger spectrum.

ET's "xylophone" design—pairing a cryogenic low-frequency interferometer with a high-power high-frequency interferometer in each arm—provides broad sensitivity from $\sim$2 Hz up to several kHz, making it well-suited for capturing both the inspiral and post-merger phases in a single observation.

4 Impact of the 20% GW Luminosity Enhancement

We now arrive at the central question: how does a 20% increase in post-merger GW luminosity change the detection prospects?

4.1 Direct Effect on SNR

Since $h \propto \sqrt{\dot{E}_{\rm GW}}$ and the total radiated energy $E_{\rm GW}$ increases by $\sim$20%, the strain amplitude of the post-merger signal increases by approximately

$$\frac{\Delta h}{h} \approx \frac{1}{2}\frac{\Delta E_{\rm GW}}{E_{\rm GW}} \approx \frac{1}{2} \times 0.20 = 0.10, \tag{6}$$

i.e., a $\sim$10% increase in strain amplitude, which translates directly to a $\sim$10% increase in SNR (since SNR $\propto h$ for matched filtering). This is a simplified estimate—the actual effect depends on how the additional energy is distributed in frequency and time. If the remnant also becomes more compact and shifts the f_2 peak (the paper reports shifts up to $\sim$100 Hz, though this is not yet confirmed to be robust against numerical resolution), the SNR gains could be somewhat larger or smaller depending on how the shifted frequency aligns with the detector's sensitivity curve.

4.2 Implications for Detection Horizon

Since SNR $\propto 1/D_L$, a 10% increase in SNR at fixed distance corresponds to a 10% increase in the maximum detectable distance ("horizon distance") at fixed SNR threshold. In terms

of the accessible *volume*, which scales as D_L^3, this yields

$$\frac{\Delta V}{V} \approx 3\frac{\Delta D_L}{D_L} \approx 3 \times 0.10 = 0.30. \tag{7}$$

A 30% increase in accessible volume means roughly 30% more detectable post-merger events per year. For a detector like CE at 40 km, where the post-merger detection horizon for optimally oriented sources is ∼100 Mpc, this pushes the horizon out to ∼110 Mpc and increases the event rate from ∼2–3 to ∼3–4 per year (using median BNS merger rate estimates). Given the large uncertainties in the BNS merger rate (which span an order of magnitude), this is a modest but non-negligible improvement.

4.3 The More Profound Implication: A New Observable

The raw detection-rate improvement, while welcome, is not the most important consequence of this work. The deeper implication is that **the post-merger GW signal carries information about neutrino flavor physics**. This opens an entirely new channel for constraining fundamental physics.

Consider what a next-generation detector would actually measure. The dominant post-merger observable is the frequency of the f_2 peak, which encodes information about the equation of state (EOS) of dense matter. Quasi-universal relations connect f_2 to the tidal deformability $\tilde{\Lambda}$ and hence to the pressure–density relationship inside neutron stars. At ET sensitivity, the f_2 frequency can be measured to $\mathcal{O}(100)$ Hz precision for sources with post-merger SNR ~ 10.

Qiu et al. report that neutrino flavor conversion can shift f_2 by up to ∼100 Hz. This is comparable to or larger than the measurement precision of next-generation detectors, meaning that **neglecting neutrino flavor transformation in waveform models could systematically bias EOS inference**. Conversely, if the EOS is independently constrained (e.g., by tidal deformability measurements from the inspiral phase, or by nuclear physics experiments), then deviations in f_2 from the expected value could serve as *evidence for or constraints on* neutrino flavor transformation in the merger environment.

This is a remarkable prospect: using a macroscopic gravitational signal emitted by a ∼20 km remnant to constrain quantum many-body neutrino physics operating on nanosecond timescales and centimeter coherence lengths.

4.4 The Overall GW Energy Budget

The 20% enhancement in total radiated GW energy also affects the *shape* of the post-merger waveform, not just its amplitude. A more bound, more compact remnant has different oscillation mode frequencies and damping times. The amplitude of specific spectral peaks, and the presence or absence of secondary peaks (such as the $f_{\rm spiral}$ mode associated with the spiral density wave pattern in the disk), can change. For detectors with sufficient SNR to resolve multiple spectral features—which will require SNR $\gtrsim$ 15–20—these differences become accessible.

Furthermore, the binding energy of the remnant is directly related to the question of whether and when it collapses to a black hole. A more bound remnant may survive longer

before collapse (or may collapse sooner if it reaches the maximum stable mass faster). The collapse event itself produces a distinctive GW signature: a quasi-normal mode ringdown at $f \gtrsim 6$ kHz. While this frequency is above the main sensitivity band of CE and ET, the *time of collapse* can be inferred from the cessation of the post-merger oscillation signal, and deviations from EOS-predicted collapse times could signal the presence of flavor transformation effects.

5 Quantitative Context: How Significant Is 20%?

To place the 20% effect in context, it is useful to compare it to other sources of uncertainty in post-merger GW predictions.

EOS uncertainty: Different equations of state predict f_2 values spanning a range of $\sim$1 kHz (roughly 2–4 kHz depending on the stiffness of the EOS). The flavor transformation-induced shift of $\sim$100 Hz is about 10% of this total range—significant but subdominant.

Numerical uncertainty: The authors estimate 10–30% numerical errors from comparisons between their moment-based (M1) transport scheme and Monte Carlo neutrino transport. The 20% effect in binding energy exceeds this uncertainty range, but the $\sim$100 Hz frequency shift does *not* survive resolution checks (the authors explicitly note they cannot rule out that the frequency shift is a numerical artifact). This is an important caveat.

Mass ratio and total mass: The simulations use only one binary configuration ($1.35 + 1.35\,M_\odot$, DD2 EOS). Different mass ratios, total masses, and EOSs will produce different quantitative results. The 20% figure should be understood as an indication of the magnitude of the effect, not a universal prediction.

Comparison to other post-merger physics: Other effects that modify the post-merger signal at comparable levels include the appearance of hyperons at high density (which can shift f_2 by several percent), phase transitions to quark matter, and magnetic field effects. Neutrino flavor transformation now joins this list of "post-merger systematics" that next-generation detectors will need to disentangle.

6 Summary and Outlook

The $\sim$20% increase in post-merger GW luminosity reported by Qiu et al. has the following consequences for Cosmic Explorer and the Einstein Telescope:

(1) Modest detection improvement: The increased GW power translates to a $\sim$10% increase in post-merger strain amplitude and a $\sim$30% increase in the volume within which post-merger signals are detectable. This is helpful—every additional detectable event is scientifically valuable—but it does not qualitatively change the detection landscape.

(2) New systematic in EOS inference: The frequency shifts and spectral modifications induced by flavor transformation are large enough to be detectable with next-generation instruments. If not accounted for in waveform models, this will bias measurements of the neutron star EOS from post-merger observations. This motivates the development of waveform models that incorporate neutrino flavor physics.

(3) A new probe of fundamental neutrino physics: Perhaps most excitingly, the sensitivity of next-generation detectors to the post-merger waveform opens the possibility of

using GW observations to constrain the physics of collective neutrino oscillations, quantum many-body effects in the neutrino sector, or even beyond-Standard-Model neutrino interactions. The post-merger signal becomes not just a probe of the EOS, but a probe of the neutrino sector—connecting particle physics to gravitational-wave astronomy in a novel way.

(4) Motivation for detector design: The importance of the 2–4 kHz band for post-merger science—now further amplified by the realization that neutrino physics imprints itself in this band—reinforces the case for the 20 km Cosmic Explorer facility with tuned high-frequency sensitivity, and for the high-frequency interferometer in ET's xylophone configuration.

The work of Qiu et al. is the first step: demonstrating that the effect exists and is significant. Future work extending to different binary configurations, equations of state, and more sophisticated flavor transformation prescriptions (e.g., based on fast-flavor instability outcomes rather than the BGK relaxation model) will be needed to fully characterize the observational signatures that CE and ET should search for.